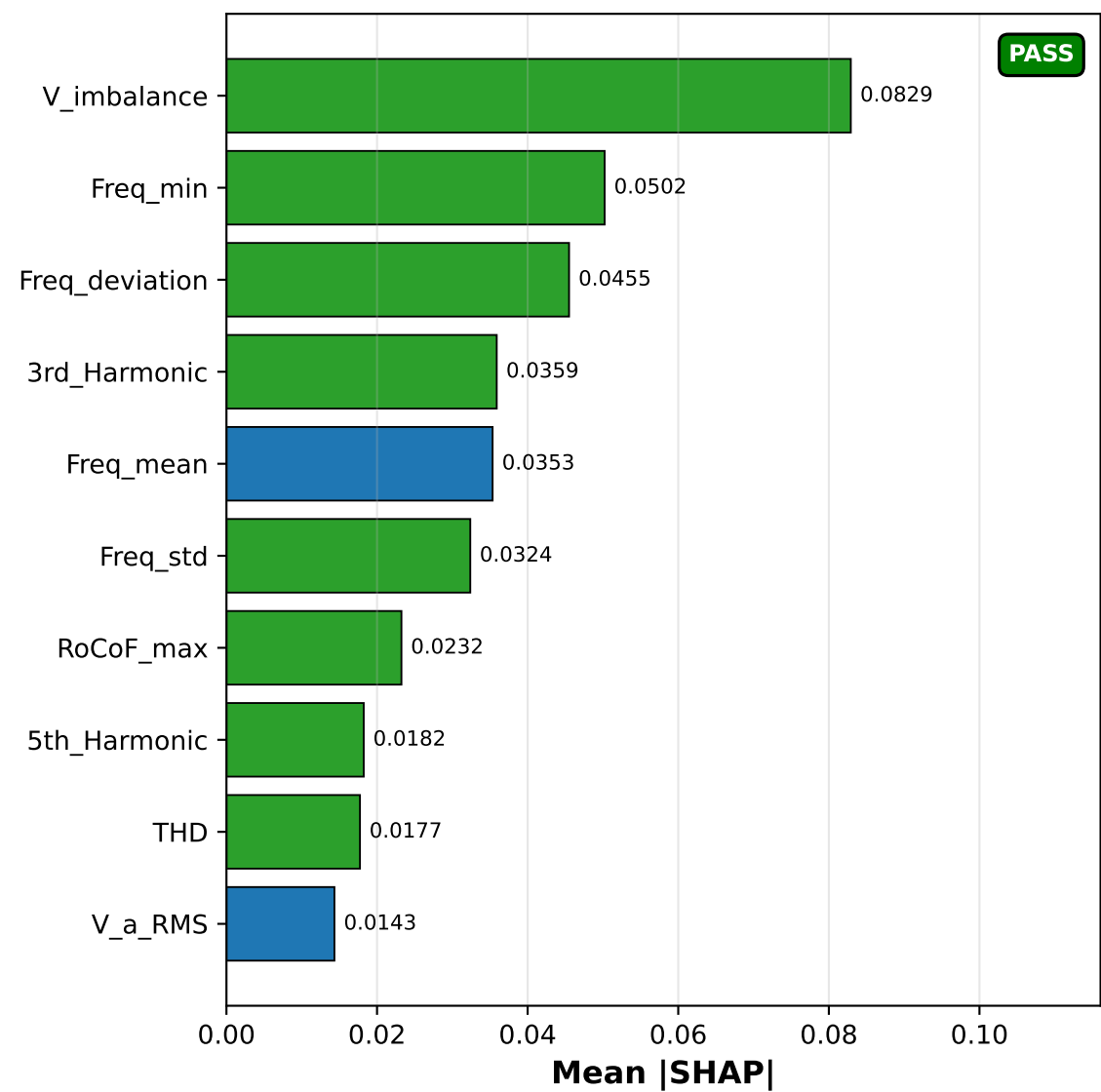
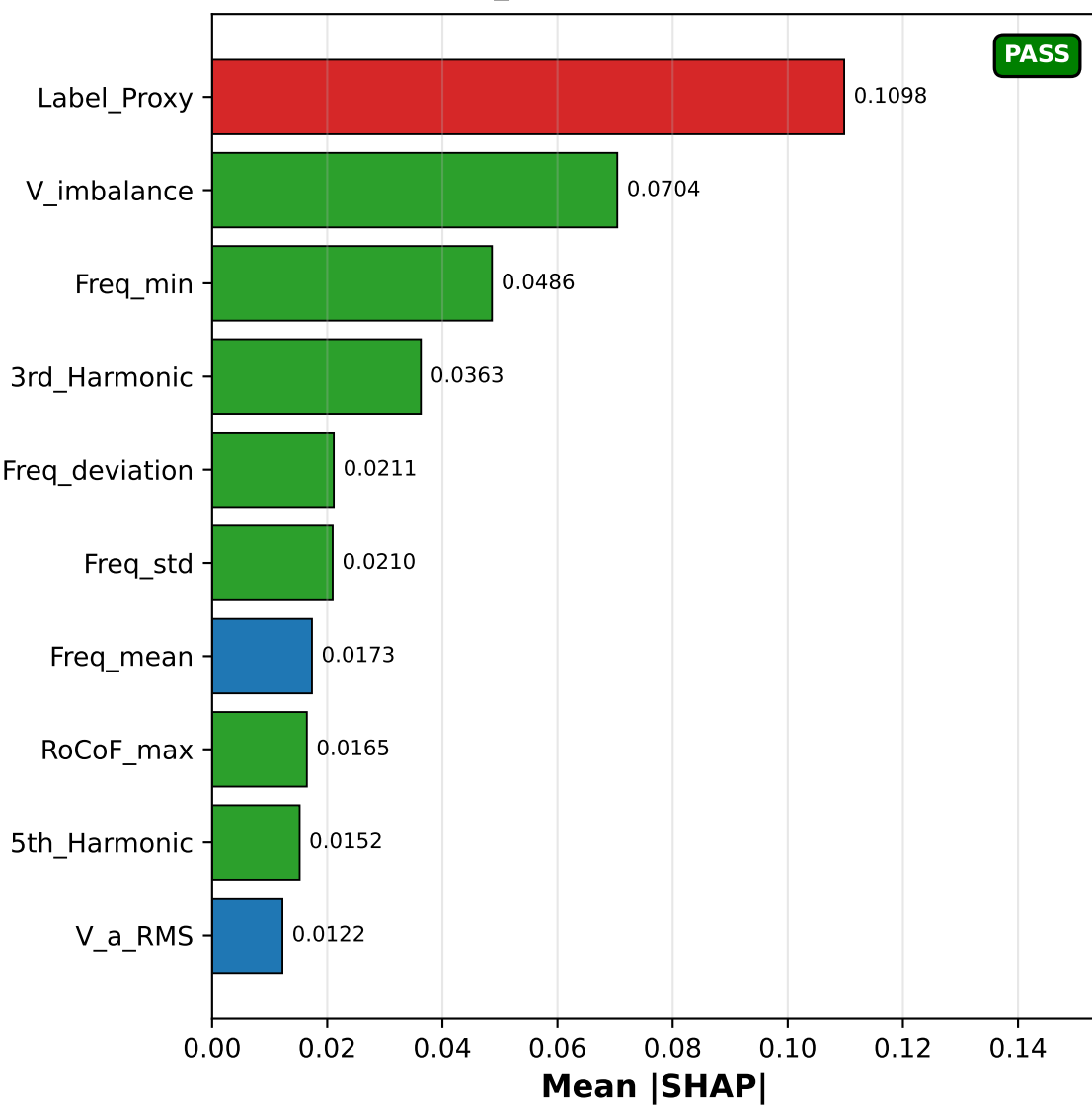


Physics-Alignment as a Falsifiable Verification Gate: SHAP Catches Spurious Correlations
All three models achieve comparable accuracy — but only the clean model passes the physics-alignment criterion.

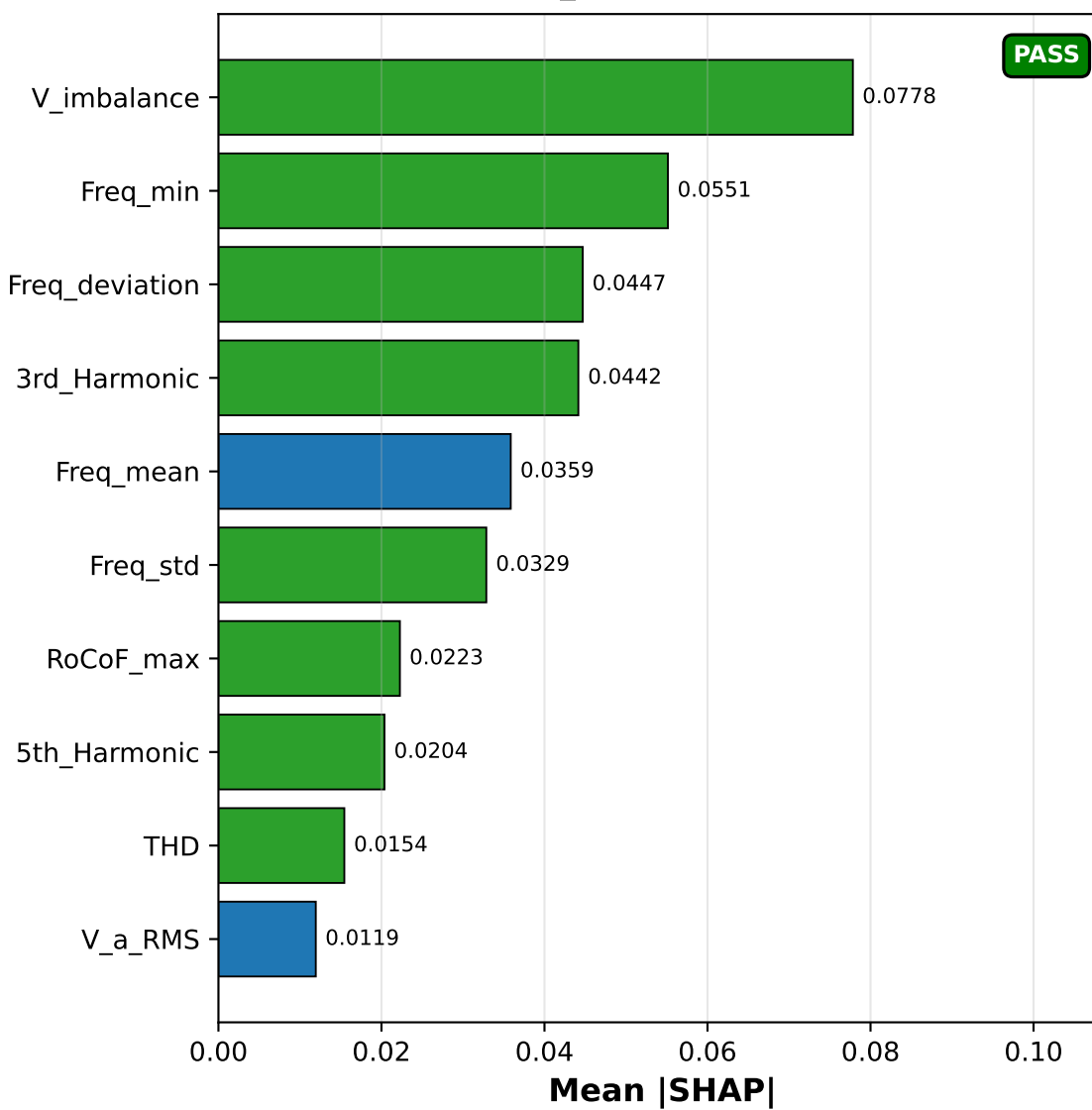
Clean Model
(15 physics features only)



Corrupted Model A
(+Label_Proxy, $r \approx 0.89$ with label)



Corrupted Model B
(+Random_Walk, $r \approx 0$ with label)



Spurious / non-causal feature Physics-canonical (expected dominant) Informative but not canonical